

Final formulation

Operational claim

In systems that exhibit a *fold-cusp-like seam* (operationally: a repeatable geometric/kinematic feature that supports a stable “rail” extraction), **rail-following signals** show a *triad of coupled signatures*:

1. **Quadratic septimal coupling (7→14) with phase closure in parity gates**
2. Extract complex analytic components $A_7(t)$ and $A_{14}(t)$ along the rail.
3. Define *parity gate windows* $\mathcal{G}_k$ by an explicit gate rule (e.g., a thresholded event in $P_7(t) = e^{i7 \arg A_7(t)}$ or an independent geometric gate detector).
4. Prediction in gates:

$$\Delta\phi_{7 \rightarrow 14}(k) = \arg(\langle A_{14} \rangle_{\mathcal{G}_k}) - 2 \arg(\langle A_7 \rangle_{\mathcal{G}_k}) \approx 0 \text{ (tight concentration).}$$

5. Amplitude relation (state-modulated):

$$\langle A_{14} \rangle_{\mathcal{G}_k} \approx C \langle A_7 \rangle_{\mathcal{G}_k}^2 E(t_k) + \eta_k,$$

where $t_k \in \mathbb{Z}_3$ is a triadic state and $E(t_k)$ takes three discrete gain levels (the “shelves”).

6. **Triadic (period-3) cyclostationary envelope (“three shelves” + sidebands)**

7. A triadic state $t_k \in \{0, 1, 2\}$ is assigned per gate (from an explicit orientation/phase estimator reduced mod $2\pi/3$).
8. In full-holonomy runs, regression of $|\langle A_{14} \rangle|$ vs $|\langle A_7 \rangle|^2$ forms **three strata** with a common slope and three intercept/gain levels.
9. Frequency-domain corollary: **triplets** at $f_g \pm f_g/3$ around the gate frequency f_g , consistent with a triadic envelope locked to gating.

10. **Coupled clock with 9- and 63-period signatures under constrained nulls**

11. Track septimal sector index $b_k \in \mathbb{Z}_7$ across gates via a pre-specified update rule $b_{k+1} = b_k + \delta_k \pmod{7}$, $\delta_k \in \{-1, +1\}$ (or explicitly justified alternatives).
12. Define $b_3(k) = b_k \pmod{3}$ and the coupled clock:

$$\tau_9(k) = t_k + 3 b_3(k) \pmod{9}, \quad z_k = \exp\left(i \frac{2\pi}{9} \tau_9(k)\right).$$

13. Prediction: the periodogram/autocorrelation of z_k (or $\tau_9(k)$) shows peaks at **9** and **63** (the LCM of 7 and 9), **and these peaks survive block-bootstrap but die under gate-order permutations that preserve marginals.**

Certification criterion (“full-holonomy-like” regime)

A run is certified *full-holonomy-like* **iff** all three diagnostics pass and all targeted falsifiers kill them:

- **Pass conditions** (pre-registered thresholds): 1) within-gate $\Delta\phi_{7 \rightarrow 14}$ concentration near a fixed angle (e.g., 0) with dispersion below threshold, 2) three-shelf stratification significant vs shelf-label shuffles, 3) 9 and 63 peaks significant vs constrained nulls.
- **Falsifiers / kill tests** (must break the certificate):
 - mistune harmonic pairing (replace 14 with nearby non-double band),
 - randomize/permute triadic labels while preserving gate timings,
 - permute gate order (preserving marginals of t_k, δ_k),
 - seam de-tilt / rail removal / orientation randomization (when meaningful).

What the “constant” becomes (tightened)

Rather than a universal constant, define a *conditional plateau parameter*:

- Choose a rail extraction $\mathcal{R}$ and preprocessing $\mathcal{P}$.
- Define a septimal order statistic $R_7(t) \in [0, 1]$ (e.g., coherence/order parameter) and compute a gate-cycle average $\bar{R}_7(n)$.
- Fit $\bar{R}_7(n)$ with a saturating model (e.g., logistic) to estimate $R_\infty(\mathcal{R}, \mathcal{P})$ with confidence intervals.

Scientific claim: within certified full-holonomy-like regimes and after controlling for SNR/gate density/bandwidth, R_∞ lies in a narrow reproducible range across domains.

Fastest path to validation

0) Freeze degrees of freedom (one-page preregistration)

Lock down *before looking at outcomes*: - rail extraction method $\mathcal{R}$, preprocessing $\mathcal{P}$, and drift correction, - gate detector (and whether it is independent of the 7/14 band used for evaluation), - exact band definitions for 7 and 14 (filters, bandwidths, edge handling), - estimators: A_7, A_{14} (analytic signal method), closure phase statistic, shelf regression method, - the coupled-clock update rule for b_k and definition of t_k , - null models and p-value construction (including how many shuffles/bootstraps), - pass/fail thresholds and multiple-testing corrections.

1) Avoid circularity (selection/evaluation separation)

Use one of: - **Split-band**: detect gates using an independent geometric feature or a different band than the one used to score coupling. - **Split-time**: choose gates on one segment, score diagnostics on a held-out segment. - **Cross-validated gating**: alternate folds for gate selection and scoring.

2) Ablation ladder (stop early if a key claim fails)

Run in this order:

1. Phase-closure test

2. Compare within-gate vs out-of-gate $\Delta\phi_{7\rightarrow 14}$ concentration.
3. Must survive block bootstrap; must fail on harmonic mistune.

4. Quadratic coupling test

5. Bicoherence/QPC at (7,7 $\rightarrow$ 14) vs constrained nulls.
6. Must fail on mistune and on gate-order permutation.

7. Three-shelf test

8. Fit $|A_{14}|$ vs $|A_7|^2$ by triadic labels t_k .
9. Must show 3 strata; must collapse under triadic label shuffles.

10. 9/63 clock test

11. Compute periodogram/autocorrelation of z_k or $\tau_9(k)$.
12. Must survive block bootstrap; must die under gate-order permutation preserving marginals.

3) Two-lab blind replication (fastest credibility boost)

- Two independent teams, same preregistered pipeline.
- Analysts blinded to condition labels (seam-present vs de-tilt/control).
- Report effect sizes + confidence intervals + null-test p-values.

4) Robustness sweep (bounded, not exploratory)

- Small, preregistered sensitivity analysis for bandwidth, gate-width, and rail jitter.
- Expect: certificate degrades gracefully outside the full-holonomy-like regime.

5) Only after success: optional addenda

Treat “prime loop” / 13–91–182 / $\sim 1/137$ as a **separate** hypothesis family with: - strict estimator lock-in, - strict multiple-testing correction, - and independent datasets.